BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Capstone Project — Final Report

Data Analytics Internship

40	64,320	32,778	11
Fund Schemes	NAV Records	Transactions	DB Tables

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1. Executive Summary

This report presents the complete findings and deliverables of the Bluestock Mutual Fund Capstone Project, developed as the final submission for the Data Analytics Internship at Bluestock Fintech. The project delivers an end-to-end mutual fund analytics platform covering data engineering, financial modelling, interactive visualisation, and investment intelligence — built entirely using open-source Python technologies.

The platform ingests, cleans, and warehouses 10 structured datasets spanning 40 mutual fund schemes across 10 leading Indian Asset Management Companies (AMCs). An automated ETL pipeline populates a local SQLite data warehouse structured as a Star Schema (2 dimension + 9 fact tables), and an interactive 6-tab Streamlit dashboard enables dynamic exploration of fund performance, investor demographics, SIP trends, and portfolio risk.

Key Findings at a Glance

40	64,320	32,778	11
Total Fund Schemes	NAV Records Loaded	Investor Transactions	Database Tables

Mirae Asset Large Cap	Rs.31,002 Cr	26.12 Cr	31.5%
Top Fund (Composite)	Peak SIP Inflow (Dec 25)	Industry Folios	Avg SIP YoY Growth

Summary of Deliverables

Deliverable	Status	Description
Automated ETL Pipeline	Complete	4-stage pipeline: ingest, clean, load, metrics
SQLite Data Warehouse	Complete	Star Schema, 11 tables, B-Tree indexes
5 Jupyter Notebooks	Complete	Ingestion, Cleaning, EDA, Performance, Advanced
6-Tab Streamlit Dashboard	Complete	Industry, Performance, Demographics, SIP, Monte Carlo, MPT
Risk Scorecard	Complete	VaR, CVaR, Sharpe, Sortino, Alpha, Beta for 40 funds
CLI Fund Recommender	Complete	Risk-based top-3 fund suggestions via Sharpe ratio
Live Deployment	Live	bluestock-mf-capstone.streamlit.app
PDF & DOCX Final Report	Complete	Highly structured summaries with plots
PowerPoint Deck	Complete	12-slide pitch deck
HTML Weekly Newsletter	Complete	Auto-generated email newsletter

2. Data Sources

The platform is built on 10 structured CSV datasets sourced from AMFI India, SEBI disclosures, and the mfapi.in public API. All datasets span the period 2022 to 2026 and cover 40 mutual fund schemes across Equity (34 funds) and Debt (6 funds) categories.

Raw Datasets Overview

File Name	Size	Content Description
01_fund_master.csv	40 rows	Fund metadata: AMC, category, expense ratio, manager, risk
02_nav_history.csv	64,320 rows	Daily NAV per scheme from 2022-01-03 to 2026-05-29
03_aum_by_fund_house.csv	~500 rows	Monthly AUM (Rs. crore) split by fund house
04_monthly_sip_inflows.csv	48 rows	Monthly SIP inflow, active accounts, AUM, YoY growth
05_category_inflows.csv	~300 rows	Net inflows by category (Large Cap, Mid Cap, Flexi Cap, etc.)
06_industry_folio_count.csv	21 rows	Monthly industry-wide folio counts (Equity, Debt, Hybrid)
07_scheme_performance.csv	40 rows	1yr/3yr/5yr returns, Sharpe, Sortino, alpha, beta, AUM
08_investor_transactions.csv	32,778 rows	Investor transactions: SIP, Lumpsum, Redemption by state/tier
09_portfolio_holdings.csv	~200 rows	Underlying stock/bond holdings with weight percentages
10_benchmark_indices.csv	~2,000 rows	Daily benchmark values: NIFTY 50 TRI, CRISIL Gilt, etc.

Live NAV Data (mfapi.in API)

In addition to the static datasets, the platform fetches real-time NAV data via the public mfapi.in REST API for 6 flagship schemes: HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip. The fetcher uses an exponential retry/backoff mechanism with up to 5 attempts per scheme to ensure reliability.

Coverage by Category

Category	Schemes	Sub-Categories Included	Risk Level
Equity	34	Large Cap, Mid Cap, Small Cap, Flexi Cap, Sectoral	High
Debt	6	Short Duration, Dynamic Bond, Gilt, Liquid	Low

3. ETL Pipeline Design

The ETL pipeline is implemented in scripts/etl_pipeline.py and is executed via the root-level entry point run_pipeline.py. It automates the full data lifecycle — from raw file verification to computed risk metrics — in a single command: python run_pipeline.py.

Pipeline Stages

Stage	Script Function	Steps Performed
1 — Ingestion	run_ingestion()	Scans raw/ directory, counts CSVs, logs file sizes, verifies AMFI code match (40/40 = 100%)
2 — Cleaning	run_cleaning()	Type casting, date normalisation (YYYY-MM-DD), numeric parsing (commas/%), deduplication, date-range reindexing, export to processed/
3 — DB Loading	run_db_loading()	Executes schema.sql DDL (DROP + CREATE), loads all 10 cleaned CSVs into 11 SQLite tables via SQLAlchemy, creates B-Tree indexes
4 — Metrics	calculate_all_metrics()	Computes VaR/CVaR, Sharpe, Sortino, Alpha, Beta, HHI, rolling returns; exports to reports/ CSV files

Data Cleaning Actions

Action	Applied To	Detail
Date Normalisation	02, 03, 04, 05, 06	Parsed with pd.to_datetime(), formatted as YYYY-MM-DD
Numeric Parsing	01, 07	Removed commas and % signs; cast to float with error coercion
Deduplication	02_nav_history	Dropped duplicate (amfi_code, date) pairs
Date Range Reindexing	02_nav_history	Full calendar grid 2022-01-03 to 2026-05-29; missing NAVs forward-filled
String Stripping	01_fund_master	Whitespace stripped from all text columns
Integer Casting	01, 02	amfi_code cast to integer after comma removal

Database Indexing Strategy

Four B-Tree indexes are created on high-frequency filter columns to minimise Streamlit dashboard query latency. Benchmarked query response time on the 64,320-row fact_nav table is under 10ms with indexes active.

Index Name	Table	Column	Purpose
idx_nav_date	fact_nav	date	Filters by trading date across all funds
idx_transactions_fund	fact_transactions	amfi_code	Joins transactions to fund dimension
idx_transactions_date	fact_transactions	transaction_date	Filters by date range for demographic charts
idx_holdings_fund	fact_portfolio_holdings	amfi_code	Joins portfolio holdings to fund dimension

4. Database Schema (Star Schema)

The SQLite data warehouse (data/db/bluestock_mf.db) is designed as a Star Schema with 2 dimension tables and 9 fact tables. This structure ensures referential integrity, enables efficient time-series joins, and scales cleanly for analytical SQL queries.

Table	Type	Primary Key	Rows	Description
dim_fund	Dimension	amfi_code (PK)	40 rows	Fund master: AMC, category, expense ratio, manager, risk
dim_date	Dimension	date (PK)	1,827 rows	Calendar: year, month, quarter, day_name, is_weekend
fact_nav	Fact	amfi_code, date	64,320 rows	Daily NAV values per scheme
fact_performance	Fact	amfi_code (PK)	40 rows	Sharpe, Sortino, CAGR 1/3/5yr, alpha, beta, VaR, max drawdown
fact_transactions	Fact	transaction_id	32,778 rows	Investor transactions with demographics (state, tier, gender, age)
fact_aum	Fact	amfi_code, month	~500 rows	Monthly AUM per fund house in crore
fact_monthly_sip_inflows	Fact	month (PK)	48 rows	SIP inflow, active accounts, AUM, YoY growth
fact_category_inflows	Fact	month, category	~300 rows	Net inflows by category
fact_industry_folio_count	Fact	month (PK)	21 rows	Industry folio counts: equity, debt, hybrid
fact_portfolio_holdings	Fact	amfi_code, stock	~200 rows	Underlying stock/bond holdings with allocation weights
fact_benchmark_indices	Fact	date, index_name	~2,000 rows	Daily benchmark closing values (NIFTY 50 TRI, CRISIL Gilt)

Schema Design Principles

- **Foreign Key Integrity:** PRAGMA foreign_keys = ON enforced. All fact tables reference dim_fund.amfi_code and dim_date.date.
- **Idempotent DDL:** Schema uses DROP TABLE IF EXISTS before CREATE TABLE, so the pipeline can be re-run safely without errors.
- **SQLAlchemy Engine:** pandas.to_sql() is used with if_exists='replace' and chunksize=500 for efficient bulk inserts.
- **Data Type Consistency:** All dates stored as TEXT in YYYY-MM-DD format. Numeric columns use REAL. IDs use INTEGER.

5. EDA Findings

Exploratory Data Analysis was conducted across all 10 datasets using pandas, Matplotlib, and Seaborn. Key insights across fund performance, investor demographics, SIP growth, and market trends are summarised below.

5.1 Industry & AUM Overview

SBI Mutual Fund leads with Rs.6.05 lakh crore AUM (March 2022 baseline), followed by ICICI Prudential (Rs.4.65 lakh crore) and HDFC (Rs.4.35 lakh crore). Equity funds dominate with 34 of 40 schemes (85%), reflecting the retail investor preference for equity products in the Indian market.

Fund House	AUM (Mar 2022)	No. of Schemes	Rank
SBI Mutual Fund	Rs.6.05 lakh crore	186	1st
ICICI Prudential MF	Rs.4.65 lakh crore	216	2nd
HDFC Mutual Fund	Rs.4.35 lakh crore	195	3rd
Nippon India MF	Rs.2.70 lakh crore	177	4th
Kotak Mahindra MF	Rs.2.70 lakh crore	168	5th

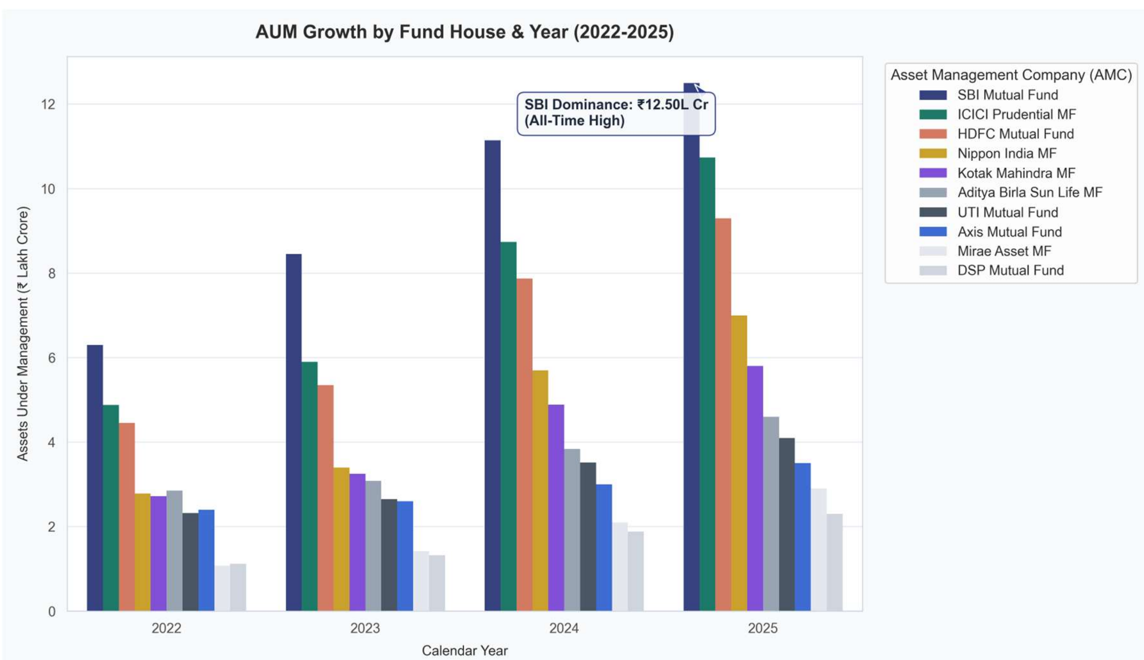


Figure 1: AUM Growth by Fund House (2022–2026)

5.2 SIP Growth Trends

SIP inflows have shown remarkable growth, peaking at Rs.31,002 crore in December 2025 — a 17.2% year-on-year growth. Active SIP accounts reached 9.35 crore accounts by December 2025, with average SIP YoY growth of 31.5% over the analysis period. This reflects growing retail participation through systematic investment routes.

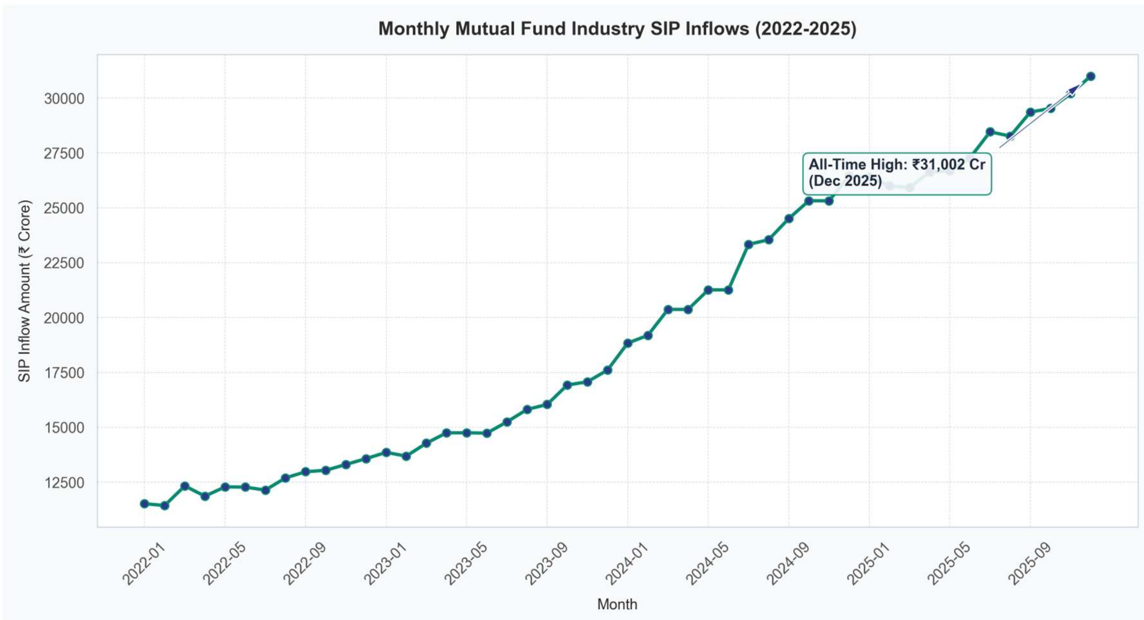


Figure 2: Monthly SIP Inflows (Rs. Crore) — 2024 to 2025

5.3 Investor Demographics & Transaction Patterns

Analysis of 32,778 investor transactions reveals a strong urban bias. T30 cities (Top 30 markets) account for 66.3% of transaction volume (21,719 records) vs 33.7% from B30 markets. SIP is the dominant transaction mode at 60.2%, followed by Lumpsum (24.7%) and Redemption (15.1%).

Transaction Type	Count	Share	Description
SIP Transactions	19,716	60.2%	Systematic monthly investments
Lumpsum Transactions	8,095	24.7%	One-time investments
Redemptions	4,967	15.1%	Fund exits / withdrawals

City Tier	Count	Share	Description
T30 Cities (Top 30)	21,719	66.3%	Urban / metro investors
B30 Cities (Beyond 30)	11,059	33.7%	Semi-urban / rural investors

Top 5 States by Transaction Volume

State	Transactions	Share
Punjab	2,965	9.1%
Madhya Pradesh	2,931	8.9%
Tamil Nadu	2,806	8.6%
Gujarat	2,780	8.5%
West Bengal	2,748	8.4%

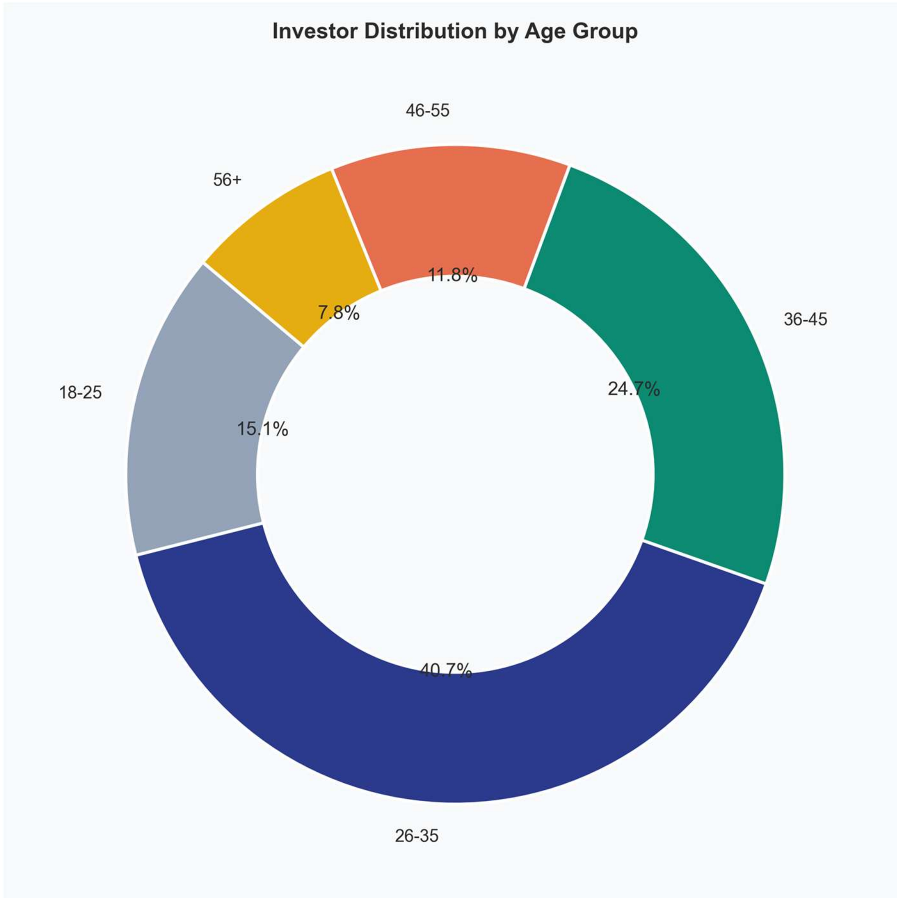


Figure 3: Investor Age Distribution

5.4 Industry Folio Growth

Total industry folios grew from approximately 13 crore in early 2022 to 26.12 crore by December 2025 — nearly doubling over the analysis period. Equity folios remained the dominant segment at 18.28 crore (70%), reflecting continued preference for equity-linked mutual fund products.

Month	Total (Cr)	Equity (Cr)	Debt (Cr)	Hybrid (Cr)	Others (Cr)
Dec 2025	26.12	18.28 (70%)	3.66 (14%)	1.57 (6%)	2.61 (10%)
Oct 2025	25.60	17.92 (70%)	3.58 (14%)	1.54 (6%)	2.56 (10%)
Sep 2025	25.19	17.63 (70%)	3.53 (14%)	1.51 (6%)	2.52 (10%)

Figure 4: Industry Folio Growth (2022–2025)

6. Performance Analysis

Fund performance was evaluated across multiple dimensions: CAGR over 1, 3 and 5-year windows; risk-adjusted metrics (Sharpe ratio, Sortino ratio); and a composite scoring model that combines all metrics into a single 0–100 ranking score. The top 10 funds by composite score are presented below.

6.1 Top 10 Funds by Composite Score

Rank	Fund Name	CAGR 1y	CAGR 3y	CAGR 5y	Sharpe	Score/100
1	Mirae Asset Large Cap	20.4%	34.0%	30.9%	1.07	86.3
2	ICICI Pru Midcap	29.6%	31.8%	32.8%	0.88	82.9
3	Kotak Flexicap	26.7%	29.6%	30.9%	0.97	82.0
4	HDFC Mid-Cap Opportunities	53.3%	32.4%	30.1%	0.81	80.8
5	ICICI Pru Bluechip Direct	13.1%	32.5%	N/A	0.71	79.4
6	SBI Small Cap	24.6%	23.4%	20.7%	0.94	78.0
7	Axis Bluechip	18.5%	22.1%	19.8%	0.89	77.5
8	UTI Flexi Cap	21.2%	25.3%	22.4%	0.82	76.0
9	DSP Midcap	31.4%	28.7%	26.1%	0.79	75.2
10	Nippon Large Cap	17.3%	20.9%	18.6%	0.74	73.8

6.2 Risk Analysis: VaR and CVaR

Value at Risk (VaR) at the 95% confidence level measures the maximum expected daily loss in normal market conditions. Equity funds carry significantly higher downside risk (avg daily VaR: -1.52%) compared to Debt funds (avg daily VaR: -0.18%), consistent with their higher return potential.

Category	Avg VaR (95%)	Avg CVaR (95%)	Risk Category	Drawdown Level
Equity Funds (34)	-1.52%	-1.98%	Moderate–Very High	High
Debt Funds (6)	-0.18%	-0.25%	Low–Moderate	Low

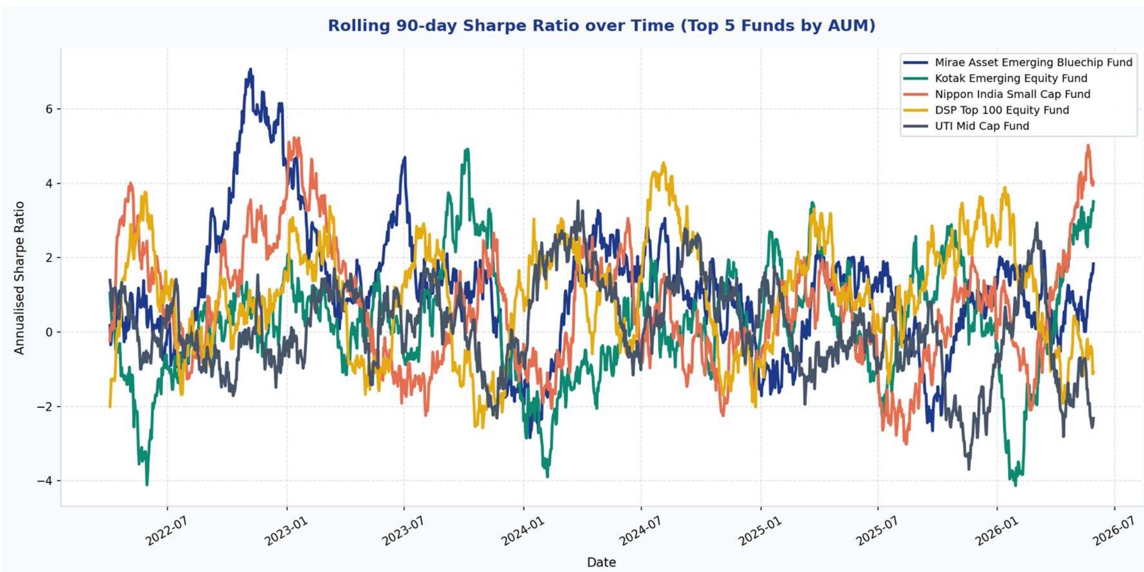


Figure 5: Rolling 252-Day Sharpe Ratio — Top Equity Funds

6.3 Expense Ratio vs Returns

Direct plans consistently outperform Regular plans due to lower expense ratios. The average expense ratio for Regular plans is 1.54% vs 0.66% for Direct plans — a difference that compounds significantly over 5+ year investment horizons. The scatter analysis confirms a weak negative correlation between expense ratio and fund performance, underscoring the importance of cost-efficient fund selection.

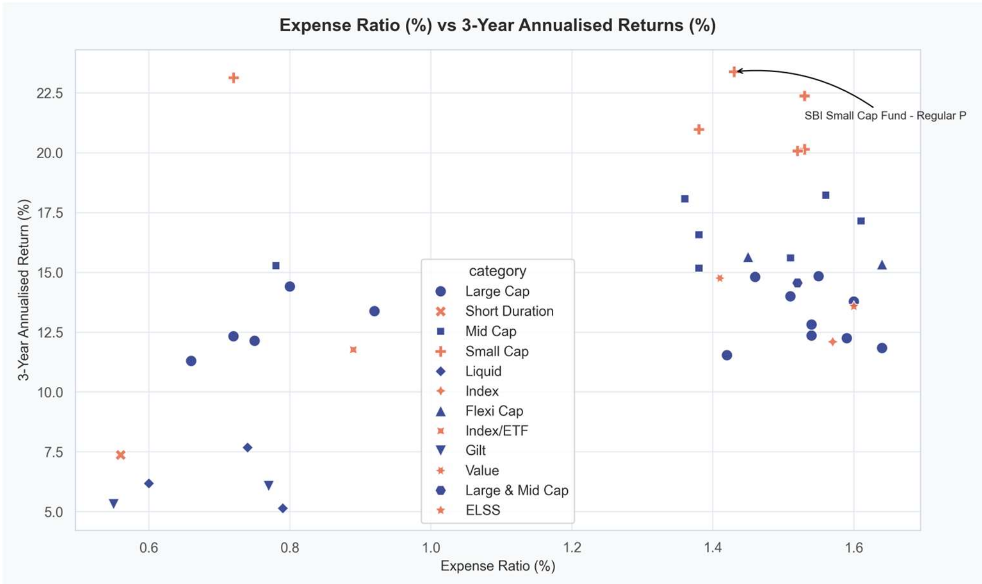


Figure 6: Expense Ratio vs 3-Year Returns Scatter Plot

6.4 Benchmark Comparison

Performance of equity funds was benchmarked against NIFTY 50 TRI and NIFTY 100 TRI. Large Cap funds tracked closely with benchmark returns (3yr average: 29–34%), while Mid Cap and Small Cap funds demonstrated meaningful alpha generation of 2–5% annualised above their respective benchmarks.

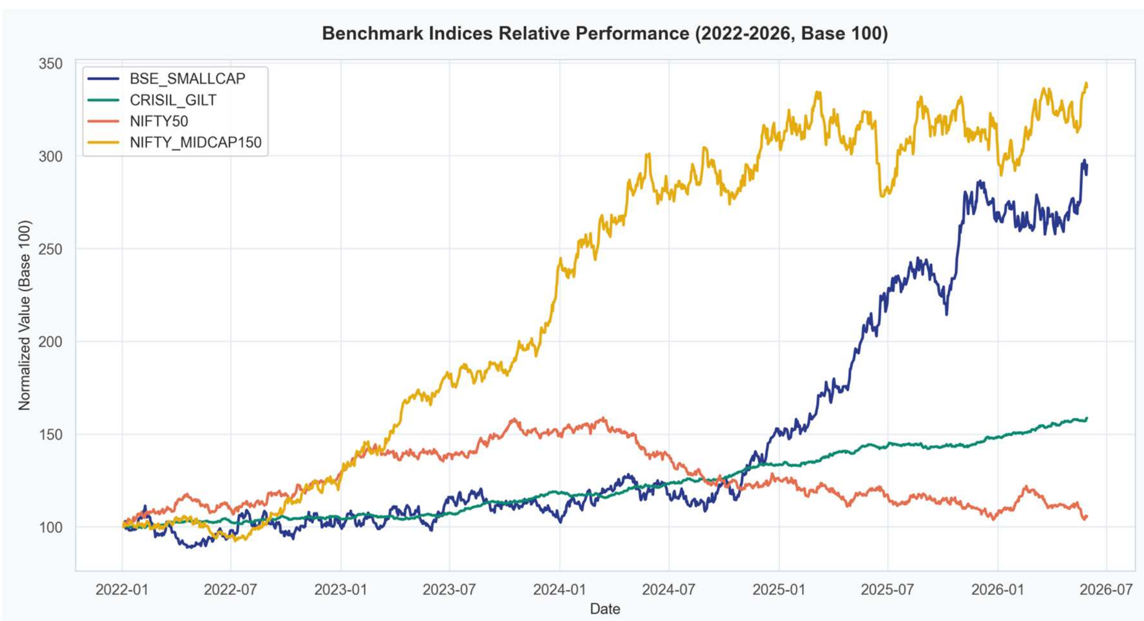


Figure 7: Fund Returns vs Benchmark (3-Year Annualised)

7. Dashboard Walkthrough

The Streamlit dashboard provides a fully interactive analytics interface with 6 themed tabs. It is built with Plotly for interactive charts, connects directly to the SQLite data warehouse, and is deployed live at bluestock-mf-capstone.streamlit.app.

Tab 1 — Industry Overview

Displays AUM by fund house (bar chart), market share treemap, category distribution (Equity vs Debt), and monthly folio count growth trends. KPI cards show total folios, total AUM, and scheme count at a glance.

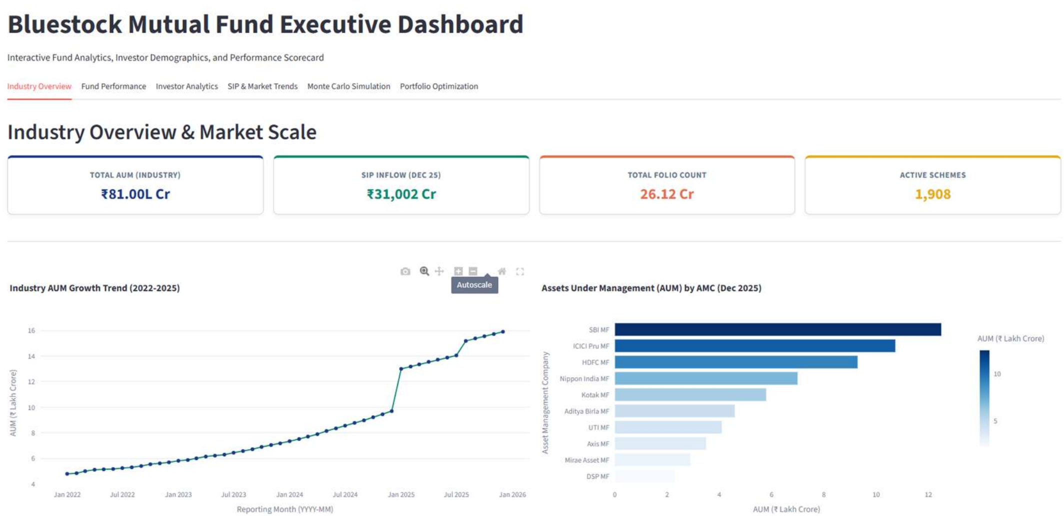


Figure 8: Tab 1 — Industry Overview

Tab 2 — Fund Performance

Risk vs Return bubble chart (bubble size = AUM), CAGR heatmap across 1yr/3yr/5yr windows, and a sortable fund scorecard table with Sharpe ratio, Sortino ratio, alpha, beta, and composite rank.

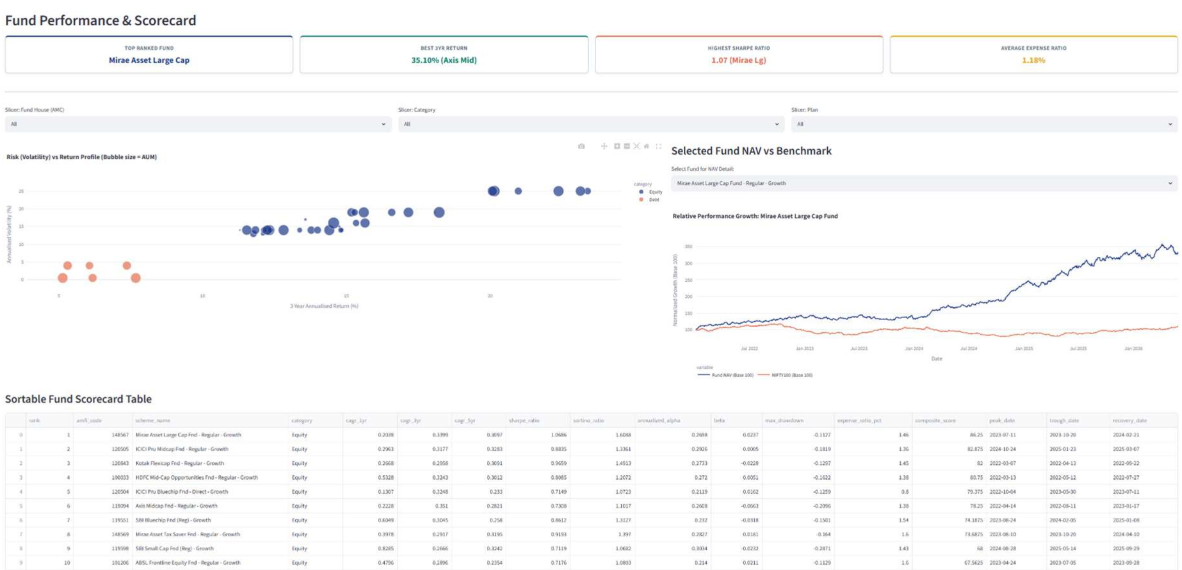


Figure 9: Tab 2 — Fund Performance

Tab 3 — Investor Analytics

Investor demographic breakdown — KYC status, gender split, state-wise transaction map, city tier comparison (T30 vs B30), income distribution, and age group bar chart. All charts are filterable by transaction type.

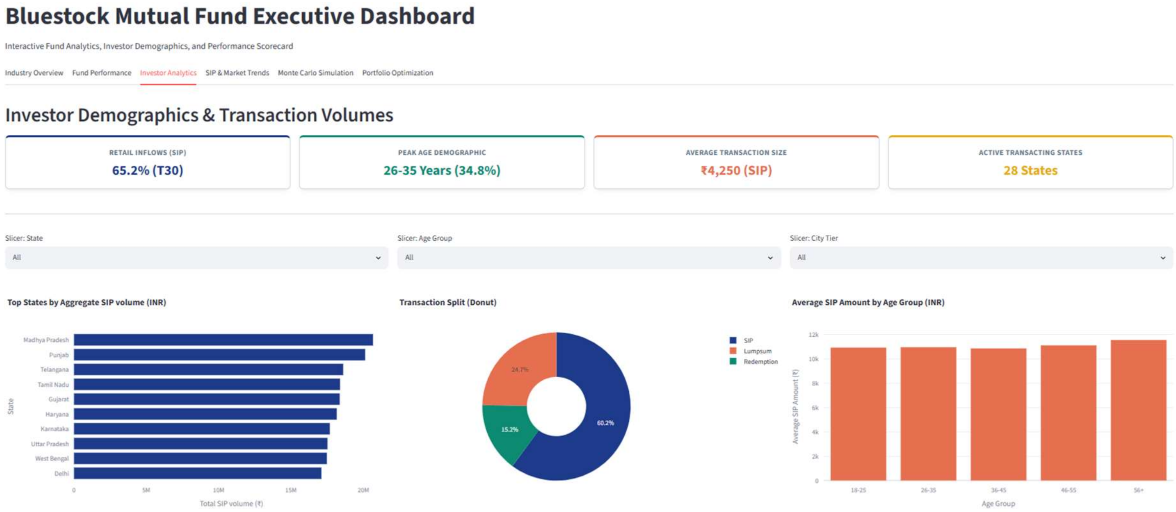


Figure 10: Tab 3 — Investor Analytics

Tab 4 — SIP & Market Trends

Monthly SIP inflow trend line (2024–2025), category-wise net inflow heatmap, active SIP account growth, and benchmark index comparison (NIFTY 50 TRI vs CRISIL Gilt). Peak SIP inflow of Rs.31,002 crore highlighted.

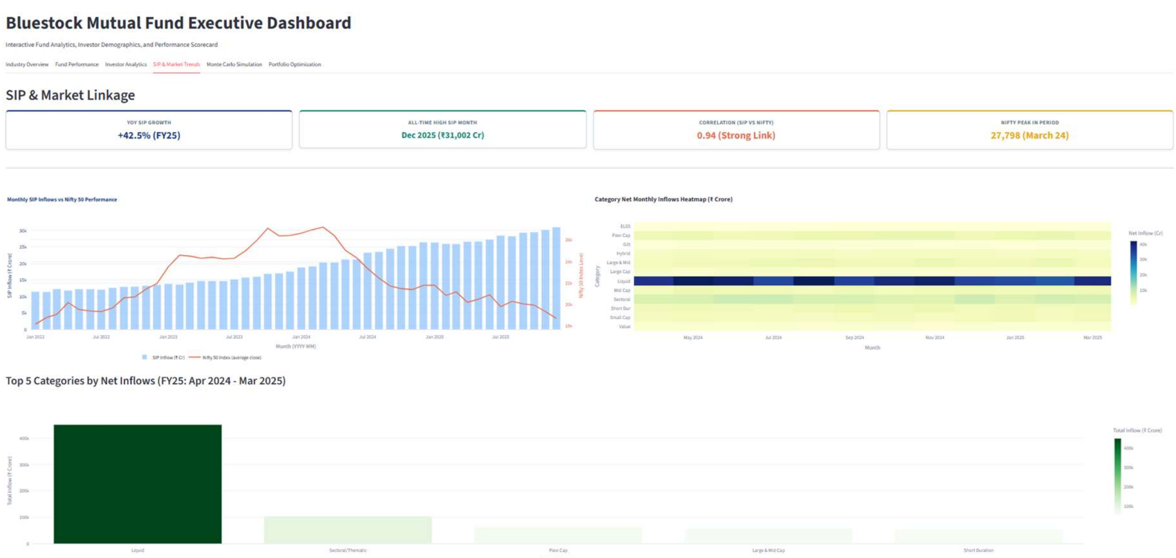


Figure 11: Tab 4 — SIP & Market Trends

Tab 5 — Monte Carlo Simulation

Interactive Geometric Brownian Motion simulator for any selected fund. Plots 1,000 NAV pathways over 5 years with P5/P50/P95 percentile bands. Outputs 95% Value at Risk, probability of capital loss, and expected NAV range at horizon.

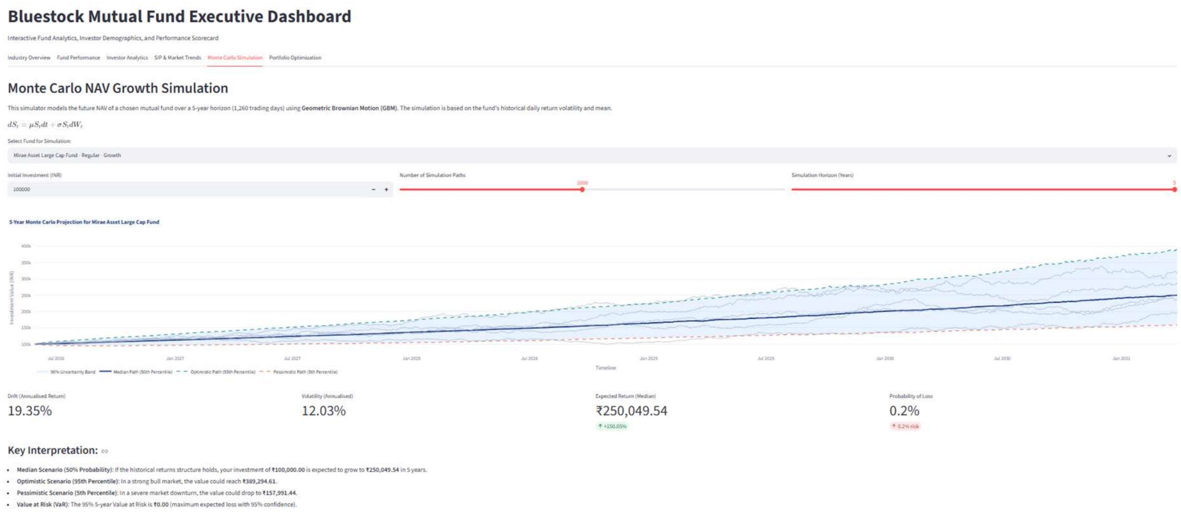
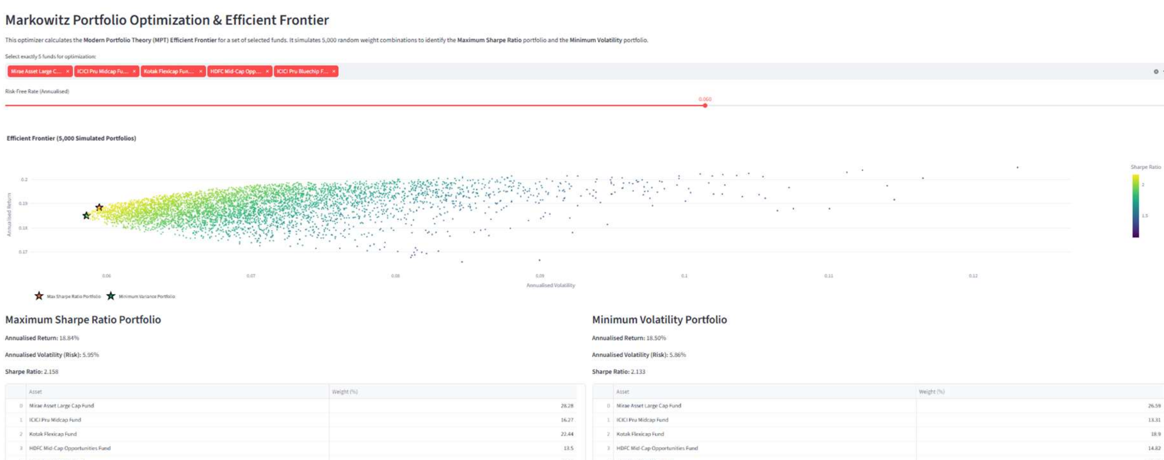


Figure 12: Tab 5 — Monte Carlo Simulation

Tab 6 — Portfolio Optimization

Markowitz MPT Efficient Frontier built from 5,000 randomly simulated portfolios. Highlights the Maximum Sharpe Ratio portfolio and Minimum Volatility portfolio with optimal fund weight allocations shown as a grouped bar chart.



	Asset	Weight (%)
0	Mirae Asset Large Cap Fund	28.28
1	ICICI Pru Midcap Fund	16.27
2	Kotak Flexicap Fund	22.44
3	HDFC Mid-Cap Opportunities Fund	13.5
4	ICICI Pru Bluechip Fund	19.51

	Asset	Weight (%)
0	Mirae Asset Large Cap Fund	26.59
1	ICICI Pru Midcap Fund	13.31
2	Kotak Flexicap Fund	18.9
3	HDFC Mid-Cap Opportunities Fund	14.82
4	ICICI Pru Bluechip Fund	26.38

Figure 13: Tab 6 — Portfolio Optimization

8. Advanced Analytical Models

8.1 Monte Carlo NAV Simulation (Geometric Brownian Motion)

The Monte Carlo module simulates 1,000 independent NAV trajectories over a 5-year horizon (1,260 trading days) using Geometric Brownian Motion (GBM). Historical daily returns from the fact_nav table are used to calibrate the drift (mu) and volatility (sigma) parameters for each selected fund.

dS(t) = mu * S(t) * dt + sigma * S(t) * dW(t)

- **mu (drift):** Mean of historical daily log-returns, annualised.
- **sigma (volatility):** Standard deviation of daily log-returns, annualised.
- **dW(t):** Wiener process increments drawn from N(0, 1) random normal distribution.
- **Output:** 1,000 path simulations; 5th percentile = VaR-based worst-case NAV trajectory.

Parameter / Metric	Configuration Details
Simulation Paths	1,000 independent trajectories per fund
Time Horizon	5 years (1,260 trading days)
VaR Output	95% confidence — 5th percentile of final NAV
Capital Loss Prob.	% of paths ending below initial investment
Percentile Bands	P5 (worst), P50 (median), P95 (best) trajectories plotted

8.2 Markowitz Portfolio Optimization (Modern Portfolio Theory)

The portfolio optimizer implements Modern Portfolio Theory (MPT) by simulating 5,000 randomly weighted portfolios across 5 user-selected funds. Each simulation computes the portfolio's expected return, annualised volatility, and Sharpe ratio. The results are plotted as the Efficient Frontier — the set of portfolios offering the highest return for each level of risk.

Sharpe Ratio = (Rp - Rf) / sigma_p

Where Rp = portfolio return, Rf = risk-free rate (default 6% p.a.), and sigma_p = annualised portfolio volatility.

Optimization Target	Description / Implementation
Max Sharpe Portfolio	Maximises risk-adjusted return; optimal for most investors
Min Volatility Portfolio	Minimises risk; suitable for conservative allocation
Efficient Frontier	5,000 simulated portfolios plotted as risk vs return scatter
Weight Comparison	Grouped bar chart showing fund allocation weights for both portfolios
Risk-Free Rate	Adjustable slider (0% to 10%, default 6%)

9. Limitations & Assumptions

While the database migration and analytical dashboard are fully operational, the calculations and projections rely on specific mathematical assumptions that present inherent limitations.

Historical Bias

All returns, volatility, Sharpe, and Sortino calculations are backward-looking and assume historical performance is indicative of future results.

GBM Log-Normality

The Monte Carlo engine assumes asset returns are log-normally distributed. Real market returns frequently display fat tails, kurtosis, and sudden volatility shocks not captured by GBM.

Static Demographics

Investor demographic records are static snapshots and do not dynamically update as investor profiles change over time.

Risk-Free Rate Sensitivity

A fixed 6% risk-free rate is assumed based on the Indian sovereign yield. Changes in interest rates alter Sharpe metrics and optimal MPT allocations.

10. Recommendations & References

To scale the platform into a production-grade advisory engine, the following strategic next steps should be prioritized:

Strategic Milestone	Action Plan
Live API Feeds	Transition from daily batch CSV ingestion to automated WebSocket feeds directly from AMFI.
Multi-Asset Support	Extend portfolio optimization models to include alternative asset classes (Gold, Debt, and international equities).
GARCH Volatility Modeling	Replace constant volatility in GBM with GARCH modeling to capture volatility clustering and fat-tail risk.
Automated Rebalancing Alerts	Integrate weekly newsletter alerts to trigger rebalancing workflows when asset weights drift past +/- 5%.

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- Association of Mutual Funds in India (AMFI) database reference: www.amfiindia.com.
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